Election Result System – Technical Approach

1. Overall System Architecture

Architecture Type: Monolithic Web Application

Frontend

- Framework: Next.js
- Responsible for:
    - Overall dashboard UI
    - Constituency result pages
    - Map visualization
    - Tables and statistics display

Backend API

- Framework: Laravel
- Responsibilities:
    - Provide API for dashboard data
    - Fetch data from Election Commission API
    - Handle admin data entry
    - Manage result updates and publishing

Database

- Engine: MySQL
- Stores:
    - Candidate master data
    - Constituency statistics
    - Vote counts
    - Referendum results
    - Postal ballot data

Cache Layer

- Cache: Redis
- Used for:
    - Dashboard statistics
    - Map data
    - Frequently accessed constituency results

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2. Overall Dashboard Implementation

Candidate Data

Backend

- Store candidate master list in `candidates` table
- API endpoint:

GET /api/candidates
GET /api/candidates/download

Frontend

- Table view
- Download as CSV/Excel

Summary Statistics

Stored Fields

Database aggregates:

total_candidates
total_voters_male
total_voters_female
total_turnout
total_complaints
complaint_status

Backend

- Aggregate query from constituency tables
- Cache results in Redis

API:

GET /api/dashboard/summary

Bangladesh Map Visualization

Map Data Source: GeoJSON of constituencies.

Frontend

- Map rendering using:
    - Leaflet or
    - Mapbox GL JS

Backend API

GET /api/dashboard/map-data

Returns:

constituency_id
party_winner
alliance_winner
postponed_status
turnout_percentage
postal_ballot_status

Map coloring based on party or alliance.

Referendum Summary

Database table:

referendum_results

Fields:

yes_votes
no_votes
valid_votes
invalid_votes
highest_range
lowest_range

API:

GET /api/dashboard/referendum-summary

Data Distribution Metrics

Calculated in backend.

Metrics:

turnout_range
close_margin_range
margin_distribution
biggest_win
lowest_win

API:

GET /api/dashboard/metrics

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3. Constituency Wise Result Page

Constituency Map View

Frontend loads selected constituency map.

API:

GET /api/constituency/{id}/map

Returns:

constituency_geojson
highlighted_area

Constituency Statistics

Database table:

constituency_results

Fields:

total_centers
total_voters
valid_votes
invalid_votes
turnout
postal_ballots
return_ballots
referendum_result
margin

API:

GET /api/constituency/{id}/statistics

Candidate Result Table

Database tables:

candidates
candidate_votes

Fields returned:

candidate_name
party
total_votes
votes_male
votes_female
age
gender

API:

GET /api/constituency/{id}/candidates

Download Constituency Result

Backend generates downloadable file.

Endpoint:

GET /api/constituency/{id}/download

Format:

- CSV
- Excel

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4. Admin Features – Technical Approach

Admin panel built in **Laravel Admin Dashboard**.

Data Source Integration (Election Commission)

Admin can configure API.

Database table:

api_sources

Fields:

api_url
api_key
last_sync_time

Backend Service:

ElectionDataSyncService

API:

POST /admin/api/sync-election-data

Flow:

Admin Clicks Sync
↓
Backend Calls Election Commission API
↓
Parse Response
↓
Update Database

Manual Data Entry

Admin forms for:

- Candidate information
- Constituency result
- Vote counts
- Turnout
- Referendum
- Postal ballots

Admin APIs:

POST /admin/candidates
POST /admin/constituency-results
POST /admin/votes

Data Editing & Correction

Admin can update records.

APIs:

PUT /admin/candidates/{id}
PUT /admin/constituency-results/{id}
PUT /admin/votes/{id}

Data Validation

Backend validation rules:

Examples:

constituency_id must be unique
votes cannot be negative
candidate must belong to constituency

Validation implemented using Laravel validation rules.

Publish Control

Results stored with status:

status = draft | published

Admin action:

Publish Result

API:

POST /admin/results/publish

Frontend dashboard only reads:

status = published

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5. Data Flow

API Flow

Election Commission API
↓
Backend Sync Service
↓
MySQL Database
↓
Redis Cache

↓
Frontend (React / Next.js)